

Let x be a random vector with $E[x] = \mu$ and $\text{cov}[x] = \Sigma$. Let A be a constant matrix and b a constant vector and $y = Ax + b$

$$E[y] = A E[x] + b$$

to find the expected value of a random vector you integrate $P(x)$ over S

since $y = Ax + b$

$$\begin{aligned} E[y] &= \int_S (Ax + b) P(x) dx \\ &= A \underbrace{\int_S x P(x) dx}_{E[x]} + b \underbrace{\int_S P(x) dx}_1 \\ &= A E[x] + b \end{aligned}$$

b) $\text{cov}[y] = A \Sigma A^T$

$$\text{cov}[y] = E[(y - E[y])(y - E[y])^T]$$

$$\begin{aligned} &= E[(Ax + b - E[Ax + b])(Ax + b - E[Ax + b])^T] \\ &\quad \begin{matrix} Ax - AE[x] & \rightarrow & A(x - E[x]) \end{matrix} \end{aligned}$$

$$\rightarrow = (E(A(x - E[x])) (A(x - E[x]))^T$$

transpose rule

$$= A \underbrace{E[(x - E[x])(x - E[x])^T]}_{\text{cov}[x]} A^T$$

$$= A \Sigma A^T \quad \text{as desired} \quad \square$$

$$2 \quad a) \quad y = \theta^T x$$

$$[\theta_0 \ \theta_1] \begin{bmatrix} 1 \\ x \end{bmatrix} \quad x_i = \begin{bmatrix} 1 \\ x_i \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

$$X^T X \theta^* = X^T Y$$

Cramer's rule

$$\begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

$$\theta_i = \frac{\det A_i}{\det A}$$

$$\theta_0 = \frac{\begin{vmatrix} 18 & 9 \\ 56 & 29 \end{vmatrix}}{\begin{vmatrix} 4 & 9 \\ 9 & 29 \end{vmatrix}} = \frac{18}{35} \quad \theta_1 = \frac{\begin{vmatrix} 4 & 18 \\ 9 & 56 \end{vmatrix}}{35} = \frac{62}{35}$$

$$y = \frac{18}{35} + \frac{62}{35}x \quad 1x$$

$$b) \quad \theta^* = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}$$

$$\frac{1}{35} \begin{bmatrix} 29 & -9 \\ -9 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} X^T Y$$

$$= \begin{bmatrix} 29 & 11 & 2 & -7 \\ -9 & 1 & 3 & 7 \end{bmatrix} \begin{bmatrix} 18 \\ 56 \\ 62 \\ 8 \end{bmatrix}$$

$\downarrow = \frac{1}{35} \begin{bmatrix} 18 \\ 62 \end{bmatrix}$ the same
as above